



Society of Petroleum Engineers

SPE-227477-MS

Rheological Insights into the Performance of High Viscosity Friction Reducers

Sherif. A. Tella, Sinopec Tech Middle East, DTV, Dhahran, Saudi Arabia; Joel. U. Gutierrez, Saudi Aramco, Dhahran, Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227477-MS](https://doi.org/10.2118/227477-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This study investigates the rheological behavior and friction reduction performance of five high-viscosity friction reducers (HVFRs) — Xanthan gum, PHPA, Guar gum, PAC-LV, and Starch — across varying concentrations and temperatures. The rheological analyses focus on shear thinning and pseudoplastic flow characteristics, providing insights into optimizing formulations for enhanced flow efficiency and drag reduction in pipeline and oilfield applications. The results show that Xanthan gum and PHPA outperform other polymers in drag reduction efficiency, with Xanthan gum achieving up to 76% drag reduction at 0.5% concentration under low-temperature conditions, and PHPA reaching 78% drag reduction at 0.1% concentration at 158°F, demonstrating superior thermal stability and consistent performance at elevated temperatures. Guar gum exhibits similar flow behavior but with reduced overall efficiency. In contrast, PAC-LV and Starch display poor friction reduction capabilities, with Starch showing negative drag reduction rates and PAC-LV suffering degradation at high temperatures. Xanthan gum demonstrates the steepest viscosity profile at 0.5% concentrations, indicating the strongest shear-thinning capacity at elevated concentrations, while PHPA shows a similar trend at low concentrations of 0.2%. At high shear rates, both polymers reach comparable viscosity plateaus, reflecting similar resistance under operational conditions. Rheological analysis highlights two key parameters: the slope of viscosity reduction and the high-shear viscosity plateau, which together determine friction reduction performance. Amplitude and frequency modulation analyses reveal the effects of polymer concentration on non-destructive deformation, highlighting that increased concentration does not necessarily enhance drag reduction. Xanthan gum and PHPA show expanded linear viscoelastic regions at higher concentrations, contributing to improved viscosity retention and proppant suspension ability in static sand-settling tests. Overall, the study identifies Xanthan gum and PHPA as optimal candidates for field applications requiring effective friction reduction, temperature resilience, and proppant transport. These findings provide valuable guidelines for selecting and formulating HVFRs to improve operational efficiency and reliability in challenging industrial environments.

Introduction

The exploration and production of unconventional hydrocarbon resources, such as shale gas and oil, have garnered significant attention worldwide as critical contributors to the global energy supply. A primary challenge associated with these reservoirs is their inherently low permeability and conductivity, which limits the natural flow of hydrocarbons. This limitation is commonly addressed through stimulation techniques such as hydraulic fracturing (Sierra et al., 2023). Fracking involves drilling into the subsurface and injecting a high-pressure fluid mixture typically composed of water, proppant (sand), and various chemical additives into the rock formation to induce fractures. These fractures enhance the connectivity of the reservoir, thereby facilitating the extraction of trapped hydrocarbons (de Melo et al., 2023). However, among the chemical additives employed in fracturing fluids, friction reducers play a pivotal role. These polymers are added to minimize frictional pressure losses between the fracturing fluid and the wellbore, enabling more efficient fluid injection and improved well performance. Proper selection and formulation of friction reducers are critical because inadequate choice can lead to substantial kinetic energy loss during fluid injection, compromising the fracturing process and the effective stimulation of the reservoir. Furthermore, improperly designed fracturing fluids may cause irreversible formation damage, resulting in diminished hydrocarbon production rates or even the premature failure of the well (Tomomewo et al., 2021).

Friction reducers are utilized in the oil and gas industry for several important reasons beyond hydraulic fracturing. These additives effectively diminish the frictional resistance between fluids and the internal surfaces of pipes during the transport of hydrocarbons, thereby enhancing flow rates and overall operational efficiency. The interaction between fluid and pipe wall friction is a critical factor in fluid transport dynamics and pipeline design. Friction arises primarily from the fluid viscosity and the interior surface roughness of the pipe. As fluid flows through a pipe, molecules adjacent to the pipe wall experience a reduction in velocity due to frictional forces at the interface with the rough surface. This velocity gradient induces shear stress, which is greatest at the pipe wall. The magnitude of this wall shear stress is influenced by factors such as fluid viscosity, flow velocity, and pipe surface roughness (Ma B., 2022). Higher viscosities and flow velocities increase shear stress, while rougher pipe surfaces further exacerbate frictional forces. Excessive friction not only impedes fluid flow but also contributes to mechanical wear and tear on pumping and transport equipment, leading to elevated maintenance costs and operational downtime. The incorporation of friction reducers mitigates these effects by lowering shear stress, thereby prolonging the service life of critical infrastructure components such as pumps and valves. This reduction in friction enhances equipment reliability and reduces overall operational costs.

Research on optimizing friction reducers for specific field conditions remains limited, with most efforts focused on maximizing friction reduction under ambient conditions. Confidentiality and lack of industry standards hinder comprehensive evaluation, though there are several tests that can be utilized to determine the performance index of friction reducers. Rheology plays a significant role in determining the effectiveness of these additives. By studying the rheological properties of the fluid, such as viscosity and shear rate, researchers can optimize the formulation of friction reducers to achieve the desired reduction in friction and increase in flow efficiency. The rheology of fracturing fluids for instance is fundamental for hydraulic fracturing design, i.e. prediction of fracture growth and geometry. Accurate measurements and good understanding of their rheological properties are essential for designing and executing an optimum treatment. The rheology of hydraulic fracturing fluids affects transport, suspension, and deposition of proppants, as well as flow back after placement (Miskimins, J., 2019). This study introduces a performance-based testing methodology to evaluate five different friction reducers based on their rheological properties characterization. The scope of this study involves conducting tests such as linear and logarithmic shear rate ramp, amplitude sweep, frequency modulation, and ramp temperature sweep measurements. The performance index of the polymers is analyzed using parameters such as viscosity, shear rate, shear stress, yield stress, temperature, elastic modulus, storage modulus, and flow transition index.

Methods and Procedures

This study introduces a performance-based methodology to evaluate the rheological behavior of five commercially available high-viscosity friction reducers (HVFRs): Xanthan gum, Guar gum, PHPA (Partially Hydrolyzed Polyacrylamide), PAC-LV (Polyanionic Cellulose – Low Viscosity), and Starch. These polymers are derived from various natural and synthetic sources, including polysaccharides, polyacrylamides, and cellulose derivatives, each with unique molecular structures and weight ranges between 0.1×10^7 g/mol and 50×10^6 g/mol. These molecular characteristics significantly impact their effectiveness in drag reduction, hydration behavior, and overall rheological performance in fracturing fluids.

The HVFRs are tested across four concentration levels: 0.1%, 0.2%, 0.5%, and 0.8%, using deionized water as the base fluid. Each solution is prepared by gradually adding the polymer to 1000 mL of deionized water under continuous mixing to ensure homogeneous dispersion. The mixture is stirred until the friction reducer is completely dissolved, with no clumps or undissolved particles. After preparation, the solution is allowed to rest briefly to achieve full uniformity. Hydration time is assessed through both visual inspection and viscosity measurement. Rheological characterization is performed using an HPHT viscometer and an HPHT acid-resistant rheometer. The study measures key parameters such as viscosity, shear rate, shear stress, yield stress, viscoelastic properties (storage and loss moduli), and flow transition index. Testing protocols include linear and logarithmic shear rate ramps, amplitude sweeps, frequency modulation, and temperature ramp measurements. These comprehensive procedures help identify the flow behavior, structural stability, and performance efficiency of each HVFR under varying operational conditions. These measurements help to identify fluid behavior under various flow conditions, allowing exploration of both low and high shear rates. They also provide insights into how polymers respond to applied stress or strain over time, revealing their time-dependent characteristics. Additionally, these measurements help to determine the transition between solid-like and liquid-like behavior and understand polymer behavior under different thermal conditions. The linear shear rate sweep is set to be 0s^{-1} to 170s^{-1} and the frequency backward sweep from 500rad/s to 0.001rad/s . In addition to flow properties, the proppant-carrying capacity of each polymer is evaluated to understand its effectiveness in hydraulic fracturing applications. Over 100 experimental runs are conducted to ensure the reliability and consistency of the results. This study provides comparative insight into the performance of different HVFRs, enabling improved selection and optimization of friction reducers tailored to specific field requirements.

Results and Discussions

The friction reducers evaluated in this study formed structured fluids whose stability is influenced by a range of interrelated factors. The rheological behavior specifically the viscosity and viscoelasticity of the continuous liquid phase plays a critical role in governing the flow characteristics and dispersion performance of these polymer-based systems. Key parameters such as polymer molecular weight, concentration, temperature sensitivity, and the interaction forces between dispersed particles and the surrounding medium significantly affect the structural integrity and stability of the fluids. Furthermore, applied shear can disrupt the delicate microstructure of these fluids, leading to a breakdown in viscosity and a loss of functional performance. Understanding these dependencies is essential for optimizing the formulation and application of friction reducers in high-shear environments, such as hydraulic fracturing operations.

Viscosity and Flow Curves

Viscosity and flow curves describe the response of polymers to varying shear rates and stresses. Viscosity, a measure of fluid resistance to flow, is influenced by molecular weight, temperature, shear stress, and shear rate. Polymers with higher molecular weights typically exhibit greater viscosity. Flow and viscosity curves illustrate the relationship between shear rate (or stress) and corresponding viscosity behavior.

The flow behavior of polymers can be classified as Newtonian, pseudoplastic, or dilatant. All polymer samples in this study exhibit pseudoplastic behavior, characterized by a decrease in viscosity with increasing shear rate commonly known as shear-thinning. This phenomenon arises from the alignment of polymer chains in the direction of shear, reducing internal resistance to flow. Figure 1 presents the viscosity and flow curves for 0.2% concentrations of Xanthan gum and PHPA in both linear and logarithmic scales. The curves clearly demonstrate pseudoplastic profiles, with structural rearrangements under shear facilitating reduced viscosity. Notably, Xanthan gum exhibits a steeper slope compared to PHPA, indicating a greater degree of shear-thinning. However, at high shear rates, both polymers reach a similar viscosity plateau, suggesting comparable flow resistance under extreme shear conditions.

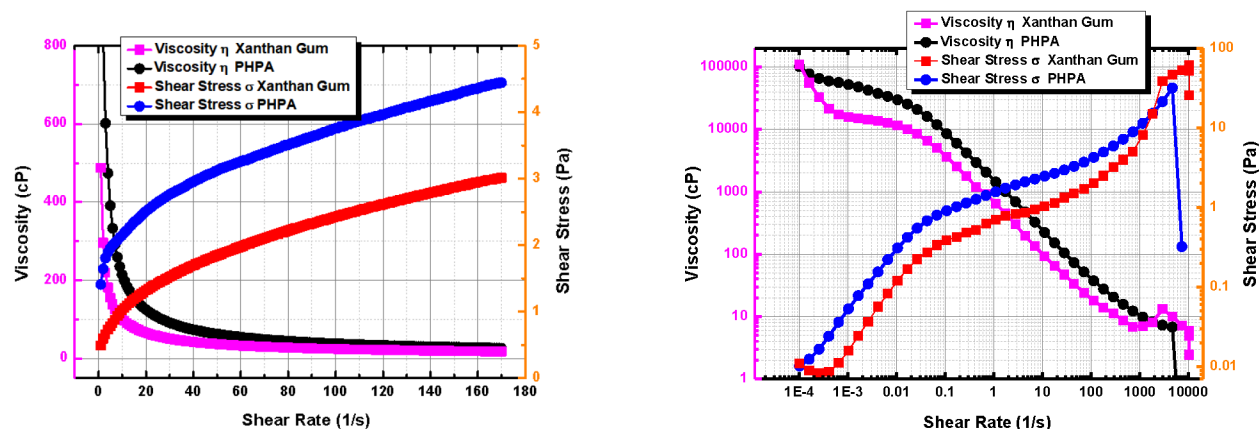


Figure 1—Viscosity and flow curves of 0.2% conc for Xanthan gum and PHPA presented on a linear and logarithm scales.

The viscosity curves for 0.2% and 0.5% concentrations of all polymers are presented in Figure 2. At 0.2% concentration, PHPA exhibits the most pronounced shear-thinning behavior, while at 0.5%, Xanthan gum demonstrates the greatest degree of shear thinning compared to the other polymers. Two critical aspects can be identified from these curves: (1) the shear-thinning slopes, which reflect the transition from low to high shear rates, and (2) the high-shear viscosity plateaus, which provide insight into the optimal viscosity range for friction reducers under operational conditions. Shear-thinning in polymers is generally attributed to structural reorientation and alignment of polymer chains when subjected to increasing shear forces, resulting in reduced resistance to flow.

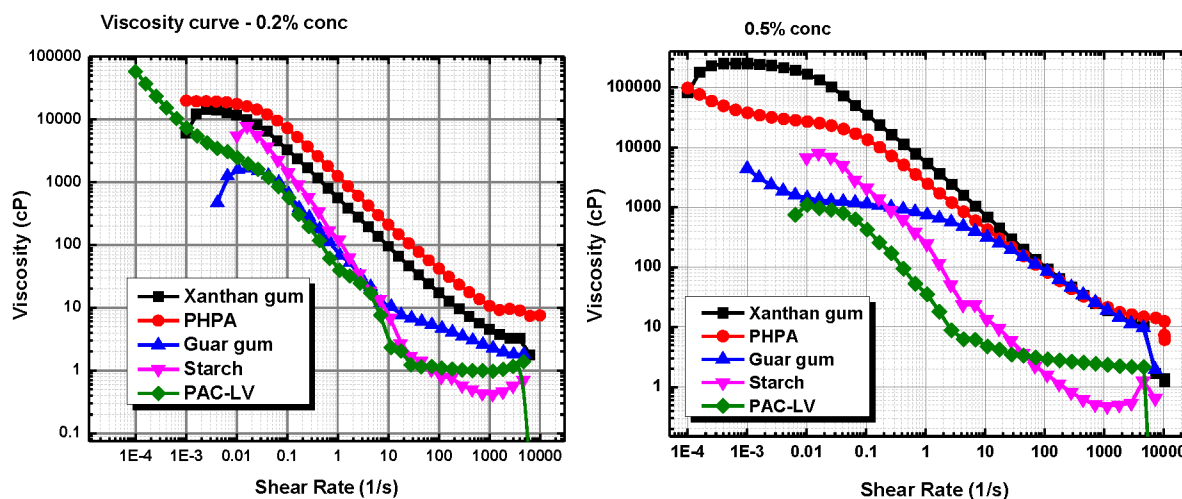


Figure 2—Viscosity curves of 0.2% and 0.5% concentrations for the polymers presented on a logarithm scale.

The viscosity curves for all the concentrations of Xanthan gum are presented in both linear and logarithmic scales as shown in Figure 3. It can be observed from Figure 3 that the high-shear viscosity plateaus vary with polymer type and concentrations, leading to distinct saturation levels. These differences are essential for determining appropriate friction reducer formulations for hydraulic fracturing applications.

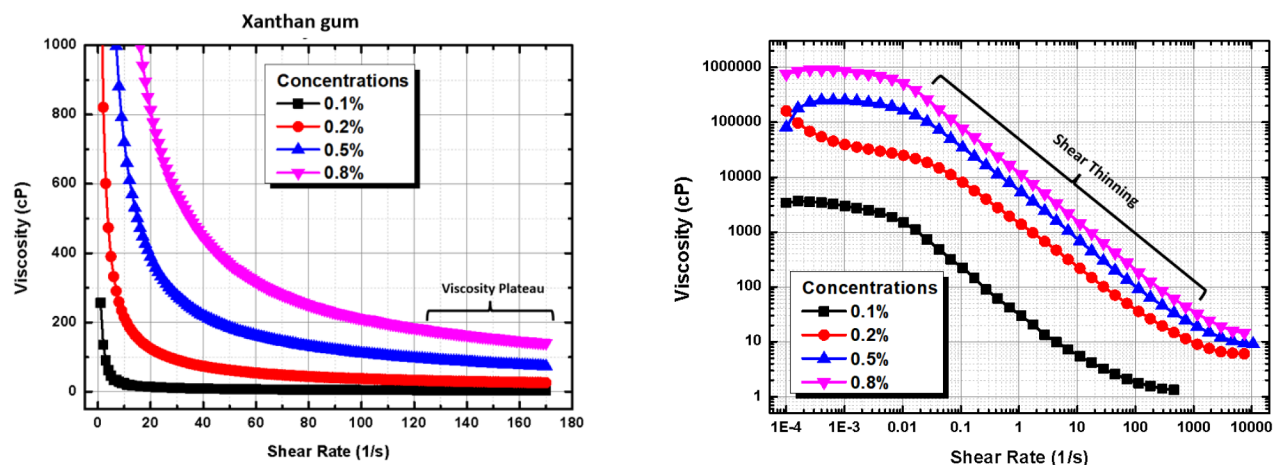


Figure 3—Viscosity curves for Xanthan gum presented on a linear and logarithm scale.

Typically, the optimal viscosity range for hydraulic fracturing fluids lies between 50 cP and 150 cP. Viscosities below 50 cP may be insufficient for effective proppant transport, while viscosities exceeding 200 cP can lead to increased pumping pressures, difficulty in fracture propagation, and potential clean-up challenges post-treatment.

Figure 4 presents the flow curves for Xanthan gum, PHPA, and Guar gum at 0.5% concentration, illustrating the behavior of these polymers under varying shear rates. These nonlinear, downward-concave curves are crucial for understanding how polymer solutions respond to deformation especially in applications like hydraulic fracturing. At low shear rates, shear stress increases gradually with increasing shear rate, while at very high shear rates, the curves tend to plateau, suggesting the polymers approach a minimum viscosity beyond which further shear may no longer reduce resistance to flow significantly. These curves allow scientists to predict performance under high shear environments like wellbores or pipelines.

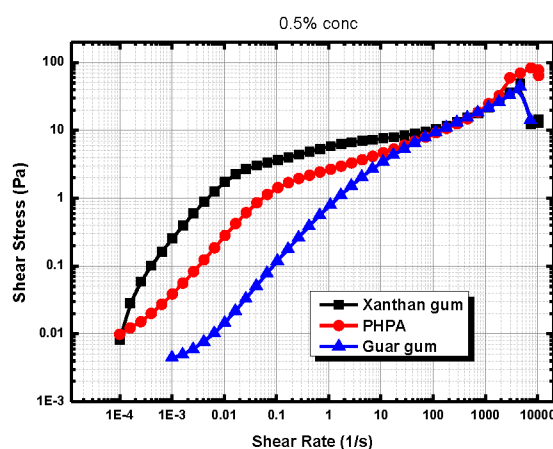


Figure 4—Flow curves for Xanthan gum, PHPA and Guar gum presented on a logarithm scale.

Frequency and Amplitude Modulation Tests

Frequency sweep modulation is a commonly used technique for analyzing the viscoelastic properties of polymers, particularly within the non-destructive deformation range to assess time-dependent behavior. In

this method, the polymer sample is exposed to a range of oscillation frequencies typically applied in a backward sweeping small-amplitude oscillatory stress or strain. The response of the polymers are then measured to extract key mechanical properties such as storage modulus, loss modulus, complex modulus, phase angle, and dynamic viscosity. This approach provides valuable insights into molecular structure, cross-link density, viscoelastic performance, and the effects of aging on mechanical characteristics of the polymers.

Figure 5 shows the backward frequency sweep profiles for 0.2% and 0.5% concentrations of Xanthan gum and PHPA. Notably, each polymer displays two intersection points between the storage and loss moduli, forming a closed region. This enclosed area represents the non-destructive deformation domain and is used to evaluate the Linear Viscoelastic (LVE) region in the amplitude modulation profiles. It can be observed that the size of this domain is smaller at lower concentrations and expands with increasing concentration, depending on the internal structure of the polymers. This suggests that higher concentrations enhance the LVE region and, correspondingly, the non-destructive deformation range.

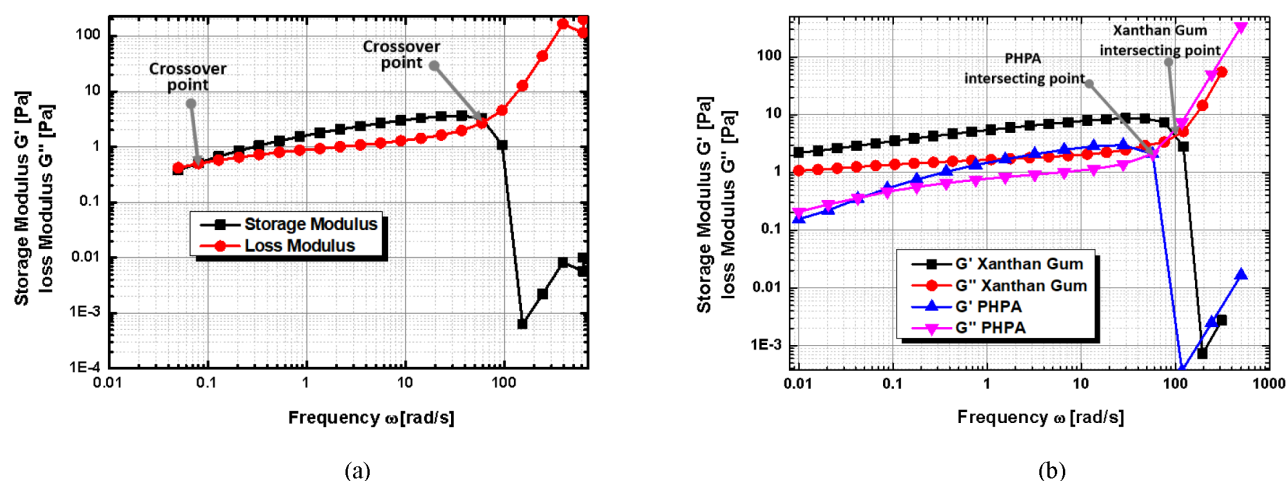


Figure 5—(a) Frequency Modulation Curves for 0.2% concentration of PHPA (b) Frequency modulation curves for 0.5% concentration of Xanthan gum and PHPA.

The amplitude sweep test is a rheological technique used to evaluate the viscoelastic properties of materials, particularly polymers. It involves applying a sinusoidal stress or strain at increasing amplitudes while maintaining a constant frequency, and measuring the response of the polymers. This test helps determine the Linear Viscoelastic Region (LVER)—the range over which the polymer exhibits predominantly elastic behavior. During the test, oscillatory shear stress or strain is gradually increased, and the resulting response is recorded as a function of amplitude. In this analysis, frequencies of 1.6 Hz (10 rad/s) and 16 Hz (100 rad/s) identified from prior frequency sweep tests were selected to represent the non-destructive and destructive deformation regimes, respectively.

Figure 6 displays the amplitude modulation curves for a 0.2% concentration of Xanthan Gum at a constant frequency of 1.6 Hz, plotted against shear stress and shear strain. The storage modulus curve is typically used to identify the constant plateau value within the Linear Viscoelastic (LVE) region, estimated at approximately 40% strain for Xanthan Gum. For the same concentration, the gel strength, yield stress, and flow stress are estimated to be 1.5 Pa, 0.9 Pa, and 1.9 Pa, respectively. Table 1 presents the corresponding values for other polymers at 0.2% concentration and 1.6 Hz. To assess viscoelastic behavior under more extreme conditions, the polymers were also evaluated at a higher frequency representative of the destructive deformation region.

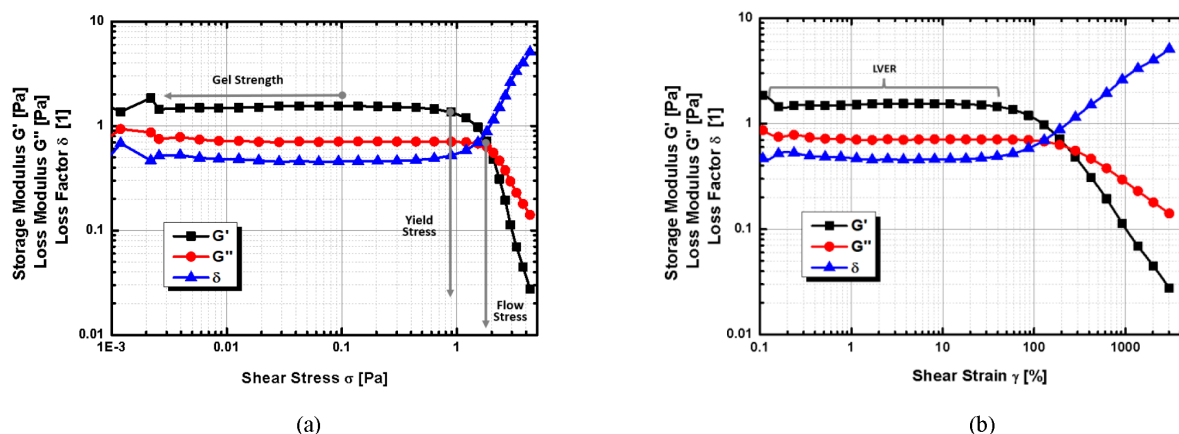


Figure 6—Amplitude modulation curves for 0.2% concentration of Xanthan Gum at frequency of 1.6 Hz (10 rad/s) presented with (a) Shear stress (b) Shear strain.

Table 1—Performance evaluation parameters for 0.1% concentrations

Parameters	Polymers - 0.1% concentrations				
	Xanthan gum	PHPA	Guar gum	Starch	PAC-LV
LVER @ 1.6Hz (10 rad/s)	20%	90%	110%	400%	200%
LVER @ 16Hz (100 rad/s)	2%	3.5%	20%	200%	0%
Gel strenght (Pa)	1	1	0.045	0.02	0.07
Yield stress (Pa)	0.2	0.7	0.05	0.085	0.1
Flow stress (Pa)	0.6	2.2	0.1	0.12	0.13
Flow Transition index	3	3.1	2	1.4	1.3

In understanding the viscoelastic behavior of the polymers under destructive deformation, amplitude modulation tests were conducted at a high frequency of 16 Hz (100 rad/s). This frequency simulates aggressive shear conditions representative of high-energy environments. Figure 7 presents the amplitude modulation curves for 0.1% concentrations of Xanthan gum and PHPA. Under this regime, both polymers exhibit measurable viscoelastic breakdown, revealing critical thresholds in their structural integrity. The Linear Viscoelastic Region (LVER) values are estimated at 2% for Xanthan gum and 3.5% for PHPA, highlighting differences in their resistance to deformation in extreme conditions.

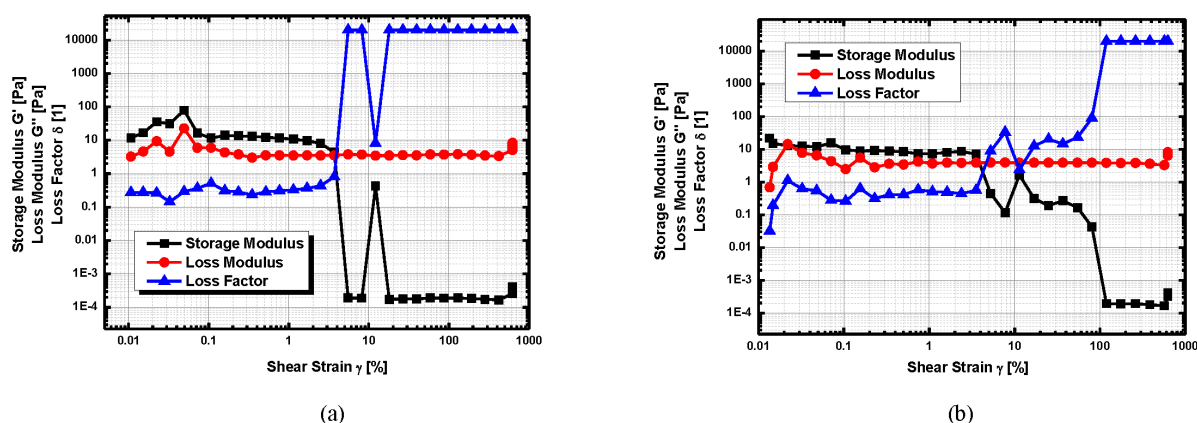


Figure 7—Amplitude curves for 0.1% concentration at frequency of 16 Hz presented with shear strain for (a) Xanthan gum (b) PHPA.

Table 2—Performance evaluation parameters for 0.2% concentrations

Parameters	Polymers - 0.2% concentrations				
	Xanthan gum	PHPA	Guar gum	Starch	PAC-LV
LVER @ 1.6Hz (10 rad/s)	40%	50%	200%	30%	100%
LVER @ 16Hz (100 rad/s)	22.5%	1%	35%	0.8%	0%
Gel strenght (Pa)	1.5	2	0.075	0.2	0.035
Yield stress (Pa)	0.9	1	0.08	0.05	0.05
Flow stress (Pa)	1.9	4	0.18	0.2	0.09
Flow Transition index	2.1	4	2.25	4	1.8

Table 3—Performance evaluation parameters for 0.5% concentrations

Parameters	Polymers - 0.5% concentrations				
	Xanthan gum	PHPA	Guar gum	Starch	PAC-LV
LVER @10 rad/s	50%	100%	100%	7%	1%
LVER @100 rad/s	50%	50%	0.05%	1%	1%
Gel strenght	6	10	0.35	2	0.025
Yield stress (Pa)	2.5	6	1	0.14	0.06
Flow stress (Pa)	8	10	0	0.5	0.095
Flow Transition index	3.2	1.7	0	3.5	1.6

Table 4—Performance evaluation parameters for 0.8% concentrations

Parameters	Polymers - 0.8% concentrations				
	Xanthan gum	PHPA	Guar gum	Starch	PAC-LV
LVER @ 1.6Hz (10 rad/s)	60%	150%	150%	10%	1%
LVER @ 16Hz (100 rad/s)	60%	70%	100%	1%	0.3%
Gel strenght	13	6.4	6.3	0.64	0.72
Yield stress (Pa)	10	7.2	3.8	0.51	0.007
Flow stress (Pa)	15	14	5.4	0.11	0
Flow Transition index	1.5	1.9	1.4	4.6	0

Viscosity-Temperature Relationship

Polymer viscosity is strongly influenced by temperature. Generally, as temperature increases, viscosity decreases due to enhanced molecular motion and weakened intermolecular forces, which reduce resistance to flow. At lower temperatures, polymer chains are more rigid and interact more strongly, resulting in higher viscosity. However, the viscosity–temperature relationship can vary based on the molecular structure of the polymer, with some showing non-linear or non-monotonic trends, especially near phase transitions. [Figure 8](#) presents viscosity–temperature curves for 0.2% and 0.5% concentrations of all polymers, revealing consistent profiles across samples: viscosity decreases with rising temperature, while saturation levels at high temperatures increase with concentration. At higher concentrations, Xanthan gum and PHPA exhibit similar temperature-dependent behavior. Conversely, at low concentrations and elevated temperatures, Starch and PAC-LV show viscosities lower than that of water. This highlights a contrasting thermal response between the two groups of polymers across different concentration levels.

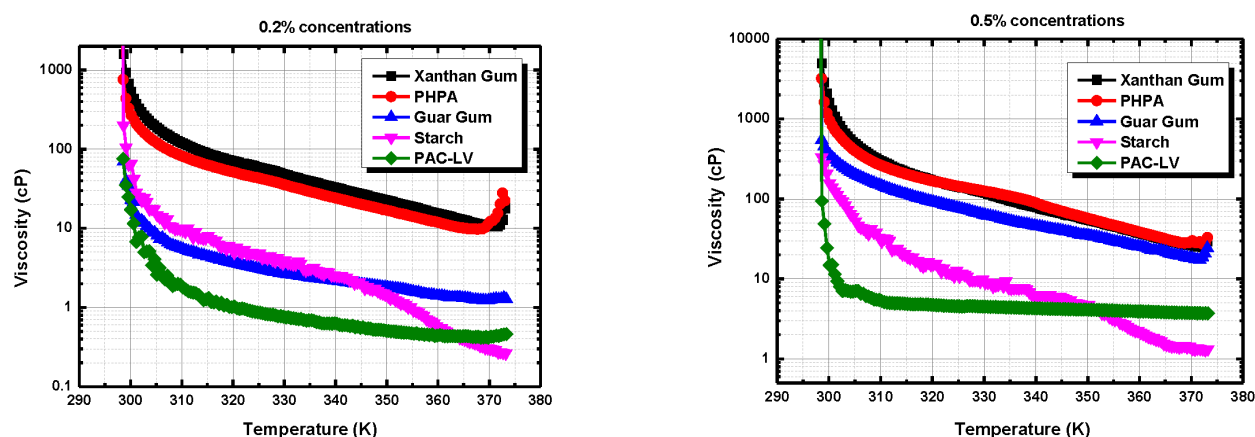


Figure 8—Viscosity-temperature relationships of all the polymers for (a) 0.2% concentration (b) 0.5% concentration.

Conclusions

Based on the comprehensive analysis of five high-viscosity friction reducers (HVFRs)—Xanthan gum, PHPA, Guar gum, PAC-LV, and Starch—the study concludes that polymer rheology is intricately influenced by concentration, temperature, and shear conditions. All polymers demonstrated pseudoplastic behavior, with flow properties improving under increasing shear rates due to molecular alignment. Xanthan gum and PHPA consistently outperformed the others in terms of shear-thinning behavior, viscosity retention, and sand-carrying capacity. Their ability to maintain structural integrity under extreme shear (above 5000%) and stress (over 20 Pa) confirms their mechanical robustness. The novel application of amplitude and frequency modulation testing in this study revealed that these two polymers exhibited expanded linear viscoelastic regions, translating to superior resistance against non-destructive deformation. Additionally, Xanthan gum and PHPA preserved their viscosities more effectively at elevated temperatures compared to PAC-LV and Starch, whose viscosities dropped below that of water. This thermal resilience is critical for applications in high-temperature environments such as oilfield operations. From a performance standpoint, drag reduction results further elevated Xanthan gum and PHPA as frontrunners, achieving up to 78% efficiency. Guar gum offered moderate benefits but lacked the robustness and thermal stability demonstrated by the leading polymers. Starch and PAC-LV, by contrast, showed poor drag reduction and compromised performance at higher temperatures.

In conclusion, the study establishes a multidimensional rheological framework to assess HVFRs and underscores the suitability of Xanthan gum and PHPA for demanding field applications requiring effective viscosity control, thermal durability, and proppant transport. These insights are pivotal for tailoring HVFR formulations to maximize operational efficiency in fluid transport and hydraulic fracturing.

Acknowledgement

This publication is based on work supported by Sinopec STME and Saudi Aramco.

References

1. Sierra, L., Eoff, L., & Palisch, T. (2023, September). Novel, Innovative Process to Improve Proppant Distribution and Improve Productivity in Hydraulically Fractured Unconventional Reservoir. In *SPE International Hydraulic Fracturing Technology Conference and Exhibition*. OnePetro. <https://doi.org/10.2118/215718-MS>
2. de Melo, R. C., Mykhaylenko, S., & Hosur, L. R. (2023, October). The Challenge to Integrate Sedimentology and Well Completion to Predict Production Performance in an

- Unconventional Reservoir. In *Abu Dhabi International Petroleum Exhibition and Conference* (p. D011S018R007). SPE. <https://doi.org/10.2118/216157-MS>
3. Tomomewo, O. S., Mann, M. D., Ellafi, A., Jabbari, H., Tang, C., Ba Geri, M., Kolawole, O., Adebisi, A., Ibikunle, O., Alamooti, M., & Iroko, A. (2021, April). Creating value for the high-saline bakken produced water by optimizing its viscoelastic properties and proppant carrying tendency with high-viscosity friction reducers. In *SPE Western Regional Meeting* (p. D021S007R005). SPE. <https://doi.org/10.2118/200809-MS>
 4. Ma, B. (2022, May). DRA Performance Predictions, Steady-State to Pseudo Steady-State Behaviors. In *PSIG Annual Meeting* (pp. PSIG-2208). PSIG.
 5. Miskimins, J. L. (Ed.). (2019). Hydraulic fracturing: fundamentals and advancements (pp. 1–795). Richardson, Texas, USA: Society of Petroleum Engineers. <https://doi.org/10.2118/9781613997192-09>